David R. Maestas, Jr.

Curriculum Vitae | November 2026

NHLBI K99/R00 Awardee · Postdoctoral Associate · University of Pittsburgh
IMMUNOENGINEERING · REGENERATIVE MEDICINE · MECHANOBIOLOGY
drm154@pitt.edu | Website | GitHub | ORCID | Google Scholar | Pitt Profile

Research Program

Understanding why injured tissues regenerate vs. scar and how to redirect their fate: My research investigates how immune-stromal interactions and mechanical cues determine tissue repair outcomes. I have developed regenerative immunotherapies and organotypic culture platforms to experimentally manipulate and longitudinally resolve these processes. My NHLBI K99/R00 research now integrates immunoengineering, vascular biomechanics, and disease-relevant models to identify therapeutic strategies that restore functional tissue remodeling.

Academic Appointments

University of Pittsburgh — *Pittsburgh, Pennsylvania* *2022–Present*
Postdoctoral Associate, Dept. of Bioengineering, Swanson School of Engineering

Current Extramural Research Support

NIH/NHLBI K99/R00 Pathway to Independence Award *2026–2031*
Principal Investigator · 1K99HL183755-01A1
Immunomechanically Tuned Vascular Grafts to Mitigate Diabetic Inflammation

NIH/NCI R25 Youth Enjoy Science (YES) Program *2026–Present*
Site Director & Co-Investigator · 2R25CA236620-07A1
UPMC Hillman Cancer Center Academy

Prior Extramural Training & Development Support

NIH/NHLBI Research Supplement to Parent R01 2023–2025
Postdoctoral Research Scholar · 3R01HL157017-04S1
Parent R01HL157017 · PI: Jonathan P. Vande Geest

National Science Foundation Graduate Research Fellow 2017–2022
Graduate Research Fellow

Education

Johns Hopkins University — *Baltimore, Maryland* *2016-2022*
Doctor of Philosophy, Biomedical Engineering
Regenerative Medicine, Fibrosis, Immunoengineering

The University of Arizona — *Tucson, Arizona* *2013-2016*
Bachelor of Science with Honors, Biomedical Engineering

Pima Community College — *Tucson, Arizona* *2010-2013*
Associate of Science with Honors
Associate of General Studies with Honors

Peer-Reviewed Publications

1. **Maestas DR Jr**, Murphy TR, Martinet KM, Moyston T, Min LX, Behrangzade A, Pemberton BJ, Joy J, Ye SH, Hussey GS, Azhar M, Wagner WR, Vande Geest JP. Resolving differential vascular graft remodeling using longitudinal multiphoton tracking in a 3D culture platform. *Materials Today Bio*. 2026;40:103626. doi: 10.1016/j.mtbio.2026.103626.

2. Martinet KM, **Maestas DR Jr**, Min LX, Gholami R, Kohyama K, Avila JL, Ye SH, Kim K, Wagner WR, Vande Geest JP. Long-term remodeling of compliance matched tissue engineered vascular grafts is sex specific. *Acta Biomaterialia*. 2026;219:341–355. doi:10.1016/j.actbio.2026.06.042.
3. Ruta A, Krishnan K, Woo J, Mejías JC, Gray-Gaillard EF, **Maestas DR Jr**, Nguyen HH, Rindone AN, Cherry C, Patatanian M, Yu FH, Yang B, Amelung C, King CD, Schilling B, Gerecht S, Fertig EJ, Davenport Huyer L, Pardoll DM, Elisseeff JH. γδ T cell-stromal networks modulate matrix composition and vascularity in foreign body response. *Nature Communications*. 2026;17(1):5334. doi:10.1038/s41467-026-71540-7.
4. Hooks JST, Yang B, **Maestas DR Jr**, Andorko JI, Ruta A, Mejias JC, Kelly SH, Hee CK, Elisseeff JH. Innate and adaptive immune responses to clinical hyaluronic acid fillers. *Journal of Cosmetic Dermatology*. 2025;24(7):e70292. doi:10.1111/jocd.70292.
5. Yang B, Rutkowski N, Ruta A, Gray-Gaillard E, **Maestas DR Jr**, Kelly SH, Krishnan K, Wu X, Wu S, Chen A, Mejías JC, Hooks JST, Vanderzee I, Mensah P, Celik N, Eric M, Abraham P, Tam A, Housseau F, Pardoll DM, Sears CL, Elisseeff JH. Murine gut microbiota dysbiosis via enteric infection modulates the foreign body response to a distal biomaterial implant. *Proceedings of the National Academy of Sciences of the United States of America*. 2025;122(20):e2422169122. doi:10.1073/pnas.2422169122.
6. Han J, Cherry C, Mejías JC, Krishnan K, Ruta A, **Maestas DR Jr**, Peña AN, Nguyen HH, Nagaraj S, Yang B, Gray-Gaillard EF, Rutkowski N, Browne M, Tam AJ, Fertig EJ, Housseau F, Ganguly S, Moore EM, Pardoll DM, Elisseeff JH. Age-associated senescent–T cell signaling promotes type 3 immunity that inhibits the biomaterial regenerative response. *Advanced Materials*. 2024;36(43):e2310476. doi:10.1002/adma.202310476.
7. Behrangzade A, Ye SH, **Maestas DR Jr**, Wagner WR, Vande Geest JP. Improving the hemocompatibility of a porohyperelastic layered vascular graft using luminal reversal microflows. *Journal of the Mechanical Behavior of Biomedical Materials*. 2024;157:106638. doi:10.1016/j.jmbbm.2024.106638.
8. Yang H, Ulge UY, Quijano-Rubio A, Bernstein ZJ, **Maestas DR Jr**, Chun JH, Wang W, Lin JX, Jude KM, Singh S, Orcutt-Jahns BT, Li P, Mou J, Chung L, Kuo YH, Ali YH, Meyer AS, Grayson WL, Heller NM, Garcia KC, Leonard WJ, Silva DA, Elisseeff JH, Baker D, Spangler JB. Design of cell-type-specific hyperstable IL-4 mimetics via modular de novo scaffolds. *Nature Chemical Biology*. 2023;19(9):1127–1137. doi:10.1038/s41589-023-01313-6.
9. Cottrill E, Pennington Z, Wolf MT, Dirckx N, Ehresman J, Perdomo-Pantoja A, Rajkovic C, Lin J, **Maestas DR Jr**, Mageau A, Lambrechts D, Stewart V, Sciubba DM, Theodore N, Elisseeff JH, Witham T. Creation and preclinical evaluation of a novel mussel-inspired, biomimetic, bioactive bone graft scaffold: direct comparison with Infuse bone graft using a rat model of spinal fusion. *Journal of Neurosurgery: Spine*. 2023;39(1):113–121. doi:10.3171/2023.2.SPINE22936.
10. Sommerfeld SD, Zhou X, Mejías JC, Oh BC, Maestas DR Jr, Furtmüller GJ, Laffont PA, Elisseeff JH, Brandacher G. Biomaterials-based immunomodulation enhances survival of murine vascularized composite allografts. Biomaterials Science. 2023;11(11):4022–4031. doi:10.1039/D2BM01845D.
11. **Maestas DR Jr**, Chung L, Han J, Wang X, Sommerfeld SD, Kelly SH, Moore E, Nguyen HH, Mejías JC, Peña AN, Zhang H, Hooks JST, Chin AF, Andorko JI, Berlinicke CA, Krishnan K, Choi Y, Anderson AE, Mahatme R, Mejia C, Eric M, Woo J, Ganguly S, Zack DJ, Zhao L, Pearce EJ, Housseau F, Pardoll DM, Elisseeff JH. Helminth egg derivatives as proregenerative immunotherapies. *Proceedings of the National Academy of Sciences of the United States of America*. 2023;120(8):e2211703120. doi:10.1073/pnas.2211703120.
12. Peña AN, Sommerfeld SD, Anderson AE, Han J, **Maestas DR Jr**, Mejias JC, Woodell-May J, King W, Ganguly S, Elisseeff JH. Autologous Protein Solution processing alters lymphoid and myeloid cell populations and modulates gene expression dependent on cell type. *Arthritis Research & Therapy*. 2022;24(1):221. doi:10.1186/s13075-022-02875-x.
13. Anderson AE, Wu I, Parrillo AJ, Wolf MT, **Maestas DR Jr**, Graham I, Tam AJ, Payne RM, Aston J, Cooney CM, Byrne P, Cooney DS, Elisseeff JH. An immunologically active, adipose-derived extracellular matrix biomaterial for soft tissue reconstruction: concept to clinical trial. *npj Regenerative Medicine*. 2022;7:6. doi:10.1038/s41536-021-00197-1.
14. Moore EM, **Maestas DR Jr**, Cherry CC, Garcia JA, Comeau HY, Davenport Huyer L, Kelly SH, Peña AN, Blosser RL, Rosson GD, Elisseeff JH. Biomaterials direct functional B cell response in a material-specific manner. *Science Advances*. 2021;7(49):eabj5830. doi:10.1126/sciadv.abj5830.
15. Cherry C, **Maestas DR Jr**, Han J, Andorko JI, Cahan P, Fertig EJ, Garmire LX, Elisseeff JH. Computational reconstruction of the signaling networks surrounding implanted biomaterials from single-cell transcriptomics. *Nature Biomedical Engineering*. 2021;5:1228-1238. doi: 10.1038/s41551-021-00770-5.

16. Wang X, Chung L, Hooks J, **Maestas DR Jr**, Lebid A, Andorko JI, Huleihel L, Chin AF, Wolf M, Remlinger NT, Stepp MA, Housseau F, Elisseeff JH. Type 2 immunity induced by bladder extracellular matrix enhances corneal wound healing. *Science Advances*. 2021;7(16):eabe2635. doi:10.1126/sciadv.abe2635.
17. Moore EM, **Maestas DR Jr**, Comeau HY, Elisseeff JH. The immune system and its contribution to variability in regenerative medicine. *Tissue Engineering Part B: Reviews*. 2021;27(1):39–47. doi:10.1089/ten.TEB.2019.0335.
18. Faust HJ, Zhang H, Han J, Wolf MT, Jeon OH, Sadtler K, Peña AN, Chung L, **Maestas DR Jr**, Tam AJ, Pardoll DM, Campisi J, Housseau F, Zhou D, Bingham CO III, Elisseeff JH. IL-17 and immunologically induced senescence regulate response to injury in osteoarthritis. *Journal of Clinical Investigation*. 2020;130(10):5493–5507. doi:10.1172/JCI134091.
19. Chung L, **Maestas DR Jr**, Lebid A, Mageau A, Rosson GD, Wu X, Wolf MT, Tam AJ, Vanderzee I, Wang X, Andorko JI, Zhang H, Narain R, Sadtler K, Fan H, Čiháková D, Jourdan Le Saux C, Housseau F, Pardoll DM, Elisseeff JH. Interleukin 17 and senescent cells regulate the foreign body response to synthetic material implants in mice and humans. *Science Translational Medicine*. 2020;12(539):eaax3799. doi:10.1126/scitranslmed.aax3799.
20. Sommerfeld SD, Cherry C, Schwab RM, Chung L, **Maestas DR Jr**, Laffont P, Stein JE, Tam A, Ganguly S, Housseau F, Taube JM, Pardoll DM, Cahan P, Elisseeff JH. Interleukin-36γ-producing macrophages drive IL-17-mediated fibrosis. *Science Immunology*. 2019;4(40):eaax4783. doi:10.1126/sciimmunol.aax4783.
21. Ardila DC, Liou JJ, **Maestas DR Jr**, Slepian MJ, Badowski M, Wagner WR, Harris D, Vande Geest JP. Surface modification of electrospun scaffolds for endothelialization of tissue-engineered vascular grafts using human cord blood-derived endothelial cells. *Journal of Clinical Medicine*. 2019;8(2):185. doi:10.3390/jcm8020185.
22. Chung L, **Maestas DR Jr**, Housseau F, Elisseeff JH. Key players in the immune response to biomaterial scaffolds for regenerative medicine. *Advanced Drug Delivery Reviews*. 2017;114:184–192. doi:10.1016/j.addr.2017.07.006.
23. Haskett DG, **Maestas DR Jr**, Howerton SJ, Smith T, Ardila DC, Doetschman T, Utzinger U, McGrath D, McIntyre JO, Vande Geest JP. 2-Photon characterization of optical proteolytic beacons for imaging changes in matrix-metalloprotease activity in a mouse model of aneurysm. *Microscopy and Microanalysis*. 2016;22(2):349–360. doi:10.1017/S1431927616000088.

Patents and Patent Applications

1. **Regenerative helminth containing compositions and uses thereof.** Elisseeff, J.H., Pardoll, D.M. and **Maestas, D.R.**, Johns Hopkins University, 2025. **U.S. Patent Application** 18/894,513.

Preprints & Manuscripts in Peer-Review

1. **Immune checkpoint inhibitors amplify type 2 immune mediated repair by pro-regenerative scaffolds.** Garcia, J., Ruta, A., Yu, F.H., Mejías, J.C., Peña, A.N., Rutkowski, N., Gray-Gaillard, E.F., Dubois, C., Cherry, C., Browne, M., Stivers, K., **Maestas, Jr., D.R.**, Krishnan, K., Bell, A.T.F., Fertig, E.J., Cooney, C.M., Cooney, D., Byrne, P., Hillel, A.T., Smith, K.N., Ji, H., Anders, R., Pardoll, D.M. and Elisseeff, J.H., *bioRxiv*, 2026. DOI: 10.64898/2026.01.31.703034
2. **Tissue Injury and Biomaterial Treatment Modulate Tumor Growth and Response to Immunotherapy.** Ruta, A., Gray-Gaillard, E.F., Garcia, J.A., Davenport Huyer, L., Mathkour, Y., Cherry, C., Patatanian, M., Mejias, J.C., **Maestas, Jr., D.R.**, Krishnan, K., Abraham, P., Wolf, M.T., Pardoll, D.M., and Elisseeff, J.H., *bioRxiv*, 2026. DOI: 10.64898/2026.02.02.703323
3. **Regulatory T cells clonally expand and contribute to stromal cell function in fibrotic response to synthetic implants.** Mejías, J.C., Krishnan, K., Ruta, A., Ramanujam, A.S., Gray-Gaillard, E.F., Huyer, L.D., **Maestas, D.R.**, Nagaraj, S., Yu, F.H., Rindone, A.N., Abraham, P., Browne, M., Fertig, E.J., Pardoll, D.M., and Elisseeff, J.H., *bioRxiv*, 2026. DOI: 10.64898/2026.01.05.697727

Research and Technology Development

- **Pro-Regenerative Immunotherapy Development | Johns Hopkins University** *2016–2022*
  Led development and investigation of a Department of Defense funded project that utilized reformulated helminth-derived immunotherapy (RSEA) that modulates innate and adaptive immune responses to enhance tissue repair. Demonstrated reduced fibrosis and enhanced functional repair outcomes across skeletal muscle, articular cartilage, and corneal injury models.
  Research output: Maestas et al., PNAS (2023); U.S. Patent Application 18/894,513.

- **Organotypic Vascular Remodeling Platform | University of Pittsburgh** *2022–2026*
  Developed a modular 3D artery-vascular graft culture platform integrating longitudinal, non-destructive multiphoton imaging and quantitative collagen microarchitecture analysis. Demonstrated sensitivity to biochemical perturbations and correspondence between selected *ex vivo* remodeling characteristics and long-term in vivo graft explants.
  Research output: Open-source model, Maestas et al., Materials Today Bio (2026).
- **FIRE Plotting Studio MATLAB Application| University of Pittsburgh** *2024–*
  Developed an open-source GUI-based MATLAB application integrating statistical analysis, gene-expression visualization, curve fitting, and publication-quality figure generation. Supports editable plots, persistent projects, statistical exports, and standalone scripts for reproducible figure recreation.
  Research output: MATLAB application; manuscript in preparation.
- **FIRE Multiphoton Workbench | University of Pittsburgh** 2024–
  Developed a GUI-based MATLAB application for multichannel, 3D microscopy visualization, image preprocessing, and quantitative second-harmonic generation (SHG) analysis of collagen architecture. Integrates image preparation, analysis, and export with saved sessions and processing provenance to support reproducible tissue-remodeling research.
  Research output: MATLAB application; manuscript in preparation.

Conference Presentations, Posters, & Presented Workshops

1. **Resolving differential vascular graft remodeling using longitudinal multiphoton tracking in a 3D culture platform, Maestas, D.R,** T.R. Murphy, K.M. Martinet, T. Moyston, L.X. Min, S. Ye, WR Wagner, JP Vande Geest. *UPPDA Research Symposium.* Poster Presentation. Pittsburgh, PA. July 2026.
2. **Vascular Engineering Workshop, Maestas, D.R.** in lieu of Vande Geest, J.P. Presented and ran a hands-on workshop focused on the design of modern cutting-edge vascular grafts. *Vasculata 2025: Vascular Bioengineering – Next Gen Systems*. Pittsburgh, PA. July 2025.
3. **2-Photon Intravital Imaging of Small Diameter Vascular Grafts in an Aortic 3D Culture System, Maestas, D.R.**, Murphy, T.R., Ye, S., Wagner, W.R., Vande Geest, J.P., NIH Supplement Scholar Workshop. Poster Presentation. October 2024.
4. **Helminth Egg Antigens as Pro-Regenerative Immunotherapies, Maestas, D.R.**, Chung, L., Sommerfeld, S.D., Pardoll, D.M., Elisseeff, J.H., *The Tumor Immunology Research in Progress. Podium Presentation, Baltimore, MD, 2021 and again in 2022*.
5. **Helminth Egg Antigens as Pro-Regenerative Immunotherapies, Maestas, D.R.**, Chung, L., Sommerfeld, S.D., Elisseeff, J.H., *BMES Conference Podium Presentation, 2021*.
6. **Skeletal Muscle Regeneration Enhanced by Natural Immunomodulatory Scaffolds, Maestas, D.R.**, Chung, L., Pardoll, D., Elisseeff, J., *BMES Podium Presentation*, Atlanta, GA, *2019*.
7. **"Tissue Engineered Vascular Grafts Seeded with Human Umbilical Cord Blood Derived Endothelial Cells," Maestas, D.R.**, Ardilla, C., Harris, D.T., Slepian, M. Vande Geest, J.P. Paper and research poster presented at The Undergraduate Research Opportunities (UROC) Consortium's Summer Colloquium; abstract published in the UROC Abstract Review, The University of Arizona, Tucson, AZ, *2015*.
8. **"Automated Whole-Body Imaging Integrated with Dermoscopy Imaging,"** Poster presented at The University of Arizona Engineering Design Day Conference, University of Arizona, Tucson, AZ, 2015.

Academic Leadership Roles

- **Site Director, UPMC Hillman Cancer Center Academy —** *2026–Present*
  Co-direct site operations for more than 10 high-school scholars across laboratories, coordinate mentors, scholar placement, progress oversight, and research presentations.
- **BioE Safety Committee Member, SSOE, University of Pittsburgh —** *2025–Present*
  Represent the Department of Bioengineering for laboratory safety planning/policy.
- **President, BME PhD Student Council, Johns Hopkins University —** *2017–2018*
  Co-led doctoral-student governance and represented trainee priorities to department
- **P-TECH Advisory Council Member & BME Recruitment Team Lead —** *2016–2020*
  Advised career-connected STEM programming, mentored participating students, and led biomedical engineering recruitment and outreach activities.
- **PhD Recruitment Leader & Big Sibling Mentor |** Hopkins BME **—** *2016–2021*
  Led and coordinated doctoral recruitment activities and peer mentorship.

Research Mentorship

- **High School STEM Research Mentorship and Supervision** 2026–
  University of Pittsburgh | Hillman Cancer Center Academy
- **Undergraduate Research Mentorship and Supervision —** *2022–Present*
  University of Pittsburgh | Johns Hopkins University
  Supervised undergraduate research contributing to the fabrication and mechanical characterization of alternative PLCL-based tissue-engineered vascular grafts for future independent investigation.
- **Laboratory Technician Intern Mentorship —** *2016–2019*
  Johns Hopkins University | University of Pittsburgh
  Mentored or supervised 7 laboratory technician interns across biomedical research and laboratory training programs.

Teaching Experience and Pedagogical Development

- **Junior Instructor (Confirmed Appointment Spring 2027)** *2027–*
  CLRES 2076: Introduction to Grant Writing for Postdoctoral Trainees
  University of Pittsburgh | School of Medicine
  Junior Instructor for a faculty-led grant-writing course for postdoctoral researchers focused on NIH grant preparation, iterative proposal development, structured peer review, and mock study-section evaluation.
- **Mathematics Instructor** *2014–2015*
  Upward Bound (TRiO), Pima Community College
  Provided mathematics instruction to high-school students participating in a federally supported college-preparatory program.
- **CRLA Level III Certified Master Tutor & Trainer** *2012–2016*
  Pima Community College
  Provided advanced STEM tutoring in mathematics, physics, and chemistry, along with academic support and tutor training.
- **STEM Teaching Courses (Audited)** *2024*
  ENGR 3002: Advanced Learning Through Evidence-Based STEM Teaching
  University of Pittsburgh | Pitt-CIRTL
  Completed non-credit coursework addressing evidence-based instructional practices, inclusive classroom strategies, inquiry- and problem-based learning, and teaching-as-research approaches to evaluating student learning.

Employment History

- **Summer Mathematics Instructor** — *Pima Community College* *2014—2015*
  Upward Bound, TRiO program. Math instructor and tutor for high school students
- **College Learning Center Tutor** — *Pima Community College* *2013—2015*
  Pima Community College East. CRLA Certified Level III Master Tutor for writing, Calculus III & Differential Equations, General & Organic Chemistry, & Physics
- **Phase II Training Store General Manager** – *Wendy's Old-Fashioned Hamburgers* *2008—2012*
  Full store management overseeing restaurant operations, food safety, cleanliness, customer service, daily operations. Phase II Training Store Certified.

Awards and Recognitions

- **Swanson School of Engineering Spotlight** — *University of Pittsburgh* *2026*
  Recognized by the Department of Engineering for leadership efforts to bring faculty, trainees, and staff together across research domains.
- **NextProf Nexus Future Faculty Workshop** — *Georgia Tech* 2023
  Georgia Institute of Technology, University of California, Berkeley, and University of Michigan. Selected for a national faculty-development program focused on the engineering faculty search, research-program development, and transition to academic independence.

- **P-TECH Outstanding Mentorship Team** — *Johns Hopkins* *2019*
  Award to recognize the largest cohort and contribution of mentors, tutors, and participants recruited to volunteer for the community P-TECH program.
- **National Science Foundation Graduate Research Fellow** — *Johns Hopkins* *2017-2022*
  The National Science Foundation. Graduate research fellowship awarded to students who demonstrate potential to strengthen the vitality of the U.S.A.'s science and engineering enterprise.
- **Drescher Graduate Medical Research Award** — *Johns Hopkins* *2017*
  Johns Hopkins University. Recognition award for outstanding first-year graduate students engaged in biomedical research.
- **Galileo Circle Scholar Student Award —** *The University of Arizona* *2016*
  A scholarship awarded to undergraduate students who demonstrate exceptional potential in the physical, mathematical, environmental, cognitive, or life sciences.
- **BIO5 Institute Innovator Award** — *The BIO5 Institute* *2016*
  The University of Arizona. Awarded at The Undergraduate and Professional Student Council Showcase. Recognizes a student exhibitor whose work most embodied multidisciplinary research, innovation in education & entrepreneurship.
- **Danny Brower Memorial Scholar** — *The University of Arizona* *2015*
  The University of Arizona. A scholarship awarded to students who demonstrated strong research experience and demonstrated financial need.
- **Highest Academic Distinction** —*The University of Arizona* *2014-2015*
  Awarded to students with more than 30 units in an academic year and attained a 4.0 grade point average during this period.
- **Dean's List with Distinction** — *The University of Arizona* *2014-2015*
  Awarded to students who attained a 4.0 grade point average.
- **Tutor of the Year-Pima East** — *Pima Community College* *2013*
  Named outstanding tutor of the year for the Pima College East Learning Center.
- **Art Alberdine Math Scholar Award** — *Pima Community College* *2012*
  A Pima Community College East Campus Scholarship award for students demonstrating high skill and potential with mathematics.
- **Board of Governor's Special Recognition** — *Pima Community College* *2012*
  A Pima Community College Special Recognition Award for students demonstrating outstanding performance at the two-year college level.
- **AMATYC Award** — *Pima Community College* *2011 & 2012*
  American Mathematical Association of Two-Year Colleges Math Scholar Award. Award for outstanding performance on the annual student mathematics league.